

Transforming Breast Cancer Diagnostics Through Deep Learning and Transparent AI Solutions

Abstract—Breast cancer is the most commonly diagnosed cancer among women worldwide and remains a major cause of mortality in the female population. This research introduces a comprehensive machine learning framework aimed at predicting the survival outcomes of breast cancer patients using data from the Surveillance, Epidemiology, and End Results (SEER) database. The dataset features two distinct outcome labels—Alive and Dead—reflecting patient survival status. A unique two-phase feature selection strategy was employed, incorporating Variance Threshold and Principal Component Analysis (PCA) to reduce data dimensionality and improve classification accuracy. All variables were standardized using the StandardScaler, and the dataset was split into training (70%; $n = 4,771$) and testing (30%; $n = 2,045$) subsets. To evaluate model performance, a LightGBM classifier was used alongside various other supervised and ensemble-based learning models, including AdaBoost, XGBoost, Gradient Boosting, and Decision Tree classifiers. Assessment of these models was carried out using both traditional train-test splits and k-fold cross-validation methods. Among all tested algorithms, the Decision Tree classifier achieved the highest accuracy of 98% across both evaluation techniques, establishing it as the most effective model for this specific dataset.

To enhance model interpretability, SHapley Additive Explanations (SHAP) were utilized to analyze the impact of each feature on model predictions. The top ten influential attributes for survival prediction included: Survival Months, Tumor Size, 6th Stage, N Stage, T Stage, Age, Estrogen Receptor Status, Progesterone Receptor Status, Regional Nodes Examined, and Regional Nodes Positive.

This study underscores the valuable role of machine learning in advancing clinical diagnostic and prognostic tools, proposing a robust and interpretable model for estimating breast cancer survival outcomes.

Index Terms—Breast Cancer, Deep Learning, Explainable AI

I. INTRODUCTION

Breast cancer is one of the most lethal cancers affecting women, but early diagnosis significantly improves prognosis. Various feature selection methods were employed across studies for breast cancer analysis. Recursive Feature Elimination, Forward Feature Selection, and correlation analyses were applied to the SEER breast cancer dataset for feature extraction [1]. Principal Component Analysis identified genomic variants in rare genetic diseases and did breast cancer data analysis.

Exploratory Data Analysis (EDA) employed Principal Component Analysis (PCA), while Receiver Operating Characteristic (ROC) curves and PCA visualized prediction capabilities across methods [2]. Random Forest (RF) combined with PCA enhanced attribute selection and diagnosis accuracy.

Recent classification approaches leveraged various AI techniques. Machine Learning and Deep Learning algorithms were

implemented for breast cancer dataset classification [3]. The SEER dataset was analyzed using Support Vector Machine (SVM), Improved Instance-Based K-Nearest Neighbour for imbalanced datasets, Random Forest, KNN, and Naive Bayes models. Multiple studies employed cross-validation with SVM and KNN, while others utilized Decision Tree, Binary SVM, AdaBoost, and catboost for cancer stage prediction[4].

Model evaluation utilized metrics including accuracy, precision, and recall. Recent studies incorporated propensity score matching for survival outcome analysis and predicted global breast cancer burden through 2040. Machine learning methods demonstrated potential in reducing false positive and negative decisions [5]. This research emphasizes SEER breast cancer dataset classification using Supervised and Ensemble Learning [1], with feature selection through Variance Threshold and PCA methods, showing a strong correlation with previously established random feature selection approaches.

We utilized the Surveillance, Epidemiology, and End Results (SEER) database for model training and validation, widely recognized as the most comprehensive and reliable cancer survivability repository. The dataset underwent rigorous pre-processing to ensure data quality, resulting in a refined dataset of 50,000 breast cancer cases, each characterized by 35 distinct features[6].

The results are a proof achieving an accuracy rate of 98%, which surpasses previously reported prediction methodologies in the literature. Moreover, the probability calculation component exhibited a mean squared error of 0.112, indicating robust reliability in probability estimation. This level of accuracy positions our model as a valuable clinical decision support tool, enabling healthcare providers to make more informed decisions regarding treatment strategies[7].

The demonstrated effectiveness of this model suggests its potential for practical implementation in clinical settings, offering physicians a reliable analytical tool for treatment planning and patient care optimization[2]. This advancement represents a significant step in applying artificial intelligence to oncological prognosis.

The early diagnosis of breast cancer significantly improves patient outcomes, making it imperative to develop accurate predictive models. Traditional machine learning models rely heavily on manual feature engineering, which can be biased and suboptimal[2]. Deep learning models, such as CNNs, automate feature extraction, uncovering complex patterns in data. However, their performance may benefit from the integration of robust classifiers like LightGBM as well as SHAP integrated with it produced great results.

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II. LITERATURE REVIEW

A. Feature Selection Methods

Effective feature selection is a cornerstone for improving model performance and interpretability in breast cancer classification tasks. Traditional methods such as *Recursive Feature Elimination (RFE)*, *Forward Feature Selection*, and *correlation analysis* have shown promise when applied to structured datasets like SEER [8]. These techniques reduce dimensionality while retaining critical information, leading to improved diagnostic performance and reduced computational overhead.

Principal Component Analysis (PCA) has emerged as a versatile tool for both dimensionality reduction and exploratory data analysis (EDA). Its integration with models such as LightGBM and Random Forest has enhanced classification robustness and attribute interpretability [9, 10]. Recent studies have also explored systematic filtering approaches like *Variance Thresholding*, which, when combined with PCA, yield a more structured and data-driven methodology compared to random feature selection techniques [11].

B. Machine Learning and Deep Learning Approaches

Mammography has long served as the standard imaging modality for breast cancer detection. However, diagnostic challenges persist—particularly in cases involving *dense parenchymal tissue* or *operator-dependent errors*—with studies reporting 10–30% of cases potentially missed during screening [12, 13, 14]. Although double-reading has improved detection sensitivity by 5–15%, machine learning offers avenues for further enhancement.

Recent ML applications have advanced the field considerably. Ensemble models combining *Random Forest*, *Naive Bayes*, and *KNN* have demonstrated superior performance in structured data scenarios, especially when addressing class imbalance using methods like *Improved Instance-Based KNN (IIBK)*. These approaches have raised benchmarks for predictive accuracy in population-level screening studies.

In parallel, *Deep Learning* models—particularly *Convolutional Neural Networks (CNNs)*—have transformed medical imaging analysis. CNNs with multi-scale kernels have shown success in automating ultrasound-based breast cancer diagnosis, while modified architectures have achieved significant breakthroughs in early-stage cancer detection and *brain tumor classification from MRI* [15]. Furthermore, *dynamic ensemble learning algorithms* are being adopted to automatically optimize DL model architectures, enhancing both diagnostic efficiency and computational scalability.

C. Explainability in Breast Cancer Prediction

The growing emphasis on *trust and transparency* in AI-driven diagnostics has catalyzed the integration of *Explainable AI (XAI)* frameworks in medical research. Tools such as *SHAP (SHapley Additive exPlanations)* have been increasingly employed to provide insight into model decision-making, addressing the "black box" nature of deep learning systems [16].

Explainability not only enhances clinician trust but also aligns predictions with domain knowledge—allowing models

to be evaluated based on medically relevant features (e.g., tumor size, grade, and shape-based attributes). The application of SHAP in conjunction with supervised and ensemble learning models has also led to improvements in model performance by revealing feature interactions that may have clinical significance.

Recent research further demonstrates how *propensity score matching* combined with survival outcome analysis can contextualize predictive performance within patient prognosis, offering a more holistic framework for evaluation [17].

D. Novel Contribution of our work

The research focuses on addressing the problem caused by a fatal disease. Despite significant advancements in breast cancer screening technologies.

A significant obstacle in the field is the opacity of current artificial intelligence (AI) systems. While recent AI advancements show promise for enhancing diagnostic precision, these systems frequently operate as "black boxes" that cannot clarify their reasoning processes. This opacity raises serious concerns in healthcare settings, where clear understanding of decision rationale is essential, as patient outcomes directly depend on timely and accurate clinical judgments.

This strategy aims to enhance the explainability of these computational approaches, which allows medical practitioners to understand the underlying logic of AI-derived recommendations, facilitates more knowledgeable and assured clinical decision-making. This research underscores the critical importance of transparent processes and interpretability in AI-based diagnostic tools, particularly regarding breast cancer detection.

This research seeks to improve the effectiveness and reliability of breast cancer detection systems in clinical settings through a hybrid XAI strategy. The overarching goal is to enhance patient care and outcomes by fostering trust and understanding between medical professionals and AI systems [18].

The development of a robust hybrid model, which integrates a Convolutional Neural Network (CNN) with the Light Gradient Boosting Model (LGBM), has led to substantial advancements in the accuracy and efficiency of breast cancer diagnosis [19]. This innovative approach leverages the strengths of both CNNs for feature extraction and LGBM's ensemble learning capabilities. The model demonstrates superior predictive performance compared to conventional methods, marking a significant milestone in breast cancer diagnostic research [20].

This study investigates a hybrid approach that incorporates a widely utilized method in explainable artificial intelligence, aimed at offering medical practitioners comprehensive explanations and critical insights [21]. These insights enhance clinical decision-making, empowering healthcare professionals to make informed choices that improve patient care and outcomes. The study implemented an explainable artificial intelligence (XAI) methodology, systematically testing it on the "SEER Breast Cancer dataset" for thorough validation. The researchers conducted an extensive evaluation to determine

the proposed system's efficacy, including comparative analysis against leading contemporary approaches [1].

The paper presents a comprehensive description of the complete system framework outlines the annotations utilized and their intended purposes the proposed model is used to generate patient predictions along with corresponding explanations. This highlights the outcomes and key findings of the study. Finally, concludes with the study's final remarks and overarching insights.

E. Explainability

The interpretability of machine learning (ML) models is increasingly recognized as a critical requirement in healthcare, where opaque decision-making can pose significant risks to patient safety. As ML applications expand into clinical domains such as cardiology, oncology, and neurology, the demand for models that offer transparent and comprehensible outcomes has become imperative [4].

To address this need, Integrated Machine Learning (IML) frameworks have gained traction, combining high predictive accuracy with interpretability. These models support clinicians in understanding the rationale behind algorithmic recommendations, thereby fostering trust and facilitating informed medical decisions. Foundational methodologies guiding the development of interpretable ML systems are well-articulated in prior work [4].

Explainable Artificial Intelligence (XAI) techniques are broadly categorized into two types: *ante-hoc* and *post-hoc* methods. Ante-hoc models are inherently interpretable by design, whereas post-hoc techniques offer retrospective insights into the decision-making of complex, often black-box, models. In clinical settings, post-hoc methods are particularly valuable because they can be applied to a wide range of pre-trained models without requiring structural modifications.

Among these, *SHAP* (*SHapley Additive exPlanations*) has emerged as a leading post-hoc explanation tool, offering mathematically grounded, visually intuitive insights into feature importance and model behavior [22]. SHAP not only quantifies the contribution of each input feature to a model's prediction but also enables clinicians to validate whether the reasoning aligns with established medical knowledge, thereby improving confidence in AI-assisted diagnostics.

F. SHapley Additive exPlanation (SHAP) (XAI)

XAI is crucial for assessing how well models perform, which helps fine-tune them for better accuracy and dependability in real-world healthcare settings. However, we must look beyond just performance metrics to understand how XAI affects trust and understanding among users. As healthcare increasingly adopts AI technologies, making these systems transparent and understandable becomes essential to build confidence among doctors and patients alike[23].

Combining different XAI methods creates significant hurdles, especially when trying to compare them or determine if findings apply across various medical scenarios. This complexity shows why we need a strong evaluation framework to thoroughly assess these explanatory models citeAdmass2024.

Creating standardized ways to measure effectiveness is vital, as this provides consistent and trustworthy evaluation methods in clinical environments. These measurements should evaluate not just how accurate and efficient the XAI tools are, but also how clear, relevant, and understandable their explanations are to healthcare professionals.

In addition to achieving high predictive performance, the ensemble model was further evaluated for explainability using SHAP (SHapley Additive exPlanations), a widely recognized framework for interpreting machine learning models[4]. SHAP provides insights into the contribution of individual features to the model's predictions, thereby enhancing its transparency and interpretability. We identified the most influential features contributing to breast cancer prediction by leveraging SHAP. Visualizations such as SHAP summary plots and dependence plots highlighted key patterns, revealing the relative importance of features extracted by the CNN and combined with original dataset attributes during LightGBM classification [4].

The SHAP analysis confirmed that certain features consistently played a pivotal role in predicting positive cancer cases, such as those related to texture, symmetry, or compactness of cell nuclei. Furthermore, SHAP's global explanations provided a comprehensive understanding of feature importance across the dataset, while local explanations illustrated the impact of specific features on individual predictions. This layer of explainability strengthens the model's utility in clinical settings, ensuring that predictions are accurate and interpretable for healthcare professionals, aiding trust and decision-making [4].

The approach introduced by Wani et. al. (2024) applies game theory principles to evaluate the contribution of each feature, or "player," towards achieving the model's objective [4]. The core premise of this method is that every feature enhances the model's predictive accuracy for a given outcome. Using the Shapley value, each feature is assigned a unique value based on its contribution to the overall reward. This concept is mathematically represented in equation:

$$g(z') = \phi_0 + \sum_{j=1}^M \phi_j z'_j$$

In this equation, g represents the interpretative model, while $z' \in \{0,1\}^M$ denotes the feature vector. M is the maximum coalition size of the vector, and $\phi_j \in \mathbb{R}$ represents the weight or Shapley value assigned to each feature j , signifying its contribution to the outcome.

The application of SHAP in healthcare testing and evaluation holds significant potential due to the accuracies it has to assess the importance of multiple biomarkers or clinical features (analogous to "players") in predicting health outcomes (the "reward"). For instance, SHAP can elucidate various medical phenomena, such as identifying factors contributing to postoperative complications or outcomes in surgical patients, as well as analyzing remote healthcare data. Additionally, SHAP is well-suited for high-dimensional datasets, including single-lead electrocardiograms (1D), photoplethysmograms (PPG), and similar types of medical data, enhancing its utility in diverse healthcare contexts. SHAP has been

widely employed by researchers to justify the outcomes of machine learning models in healthcare. Wani et. al. (2024) used SHAP to explain the expected results of their approach for infants with pneumonia [4]. Similarly, Wang et. al. (2024) leveraged SHAP to elucidate their predictions in research focused on anticipating adverse surgical outcomes [24].

To identify breast cancer patients they would develop lymph node metastases following neoadjuvant therapy, Agarwal et. al. (2024) developed machine learning models and employed SHAP to interpret their results [25]. They demonstrated that interpretable tree-based algorithms, leveraging patient and tumor characteristics from pre-operative procedures, could achieve high precision in forecasting axillary lymph node metastasis. In a similar approach, Deepak and colleagues (2024) developed an explainable framework for detecting breast cancer from ultrasound images, incorporating texture analysis to build and train a decision tree ensemble model. Their methodology employed SHAP values to generate comprehensive explanations by examining specific decision trees and their test conditions, thereby significantly improving the transparency of the diagnostic results[26].

[2]worked on cvd and used XG boost as well as SHAP . This approach provided insights into key risk factors and delivered reliable predictions despite data imbalance and low CVD prevalence. These examples highlight SHAP's versatility and effectiveness in enhancing healthcare models' transparency and interpretability. Wang et. al. (2024) developed a machine learning-based model to predict breast cancer patient survival using data from the Netherlands Cancer Registry [24]. For interpreting the model outcomes, the researchers implemented both Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) techniques. Their analysis revealed strong agreement between these two methodologies when explaining how different features contributed to the model's predictions.

In an effort to provide additional support to healthcare professionals, Karthiga et al. (2024) developed an auxiliary system that combines ensemble learning with reasoning techniques to forecast breast cancer recurrence. By integrating SHAP values into the framework, the system offered individualized explanations for each prediction, allowing clinicians to assess the reliability of the model and make more informed clinical judgments. This methodology significantly improved the interpretability of the system, effectively connecting machine-generated insights with real-world clinical practice.

SHAP values (Figure 1) indicate feature significance and their effect on model predictions. The color scale ranges from blue (low values) to pink (high values), with the x-axis position showing the prediction impact. Features appear in descending order of importance, with Survival Months demonstrating the greatest influence on prediction outcomes[5].

The SHAP Summary Plot (Figure 2) illustrates how features affect model predictions, with each dot representing an individual prediction[27]. The dot color indicates the feature value (blue for lower values, pink for higher values). The horizontal position shows the SHAP value (prediction impact), while the vertical axis lists features ranked by importance.

Survival Months displays the widest range of effects, followed by Tumor Size Log, whereas the 6th Stage shows effects mostly clustered around zero.

The SHAP feature importance plot organizes the input variables based on their average absolute SHAP values. Among all features, Survival Months exhibits the greatest influence on the model's predictions, with an average contribution of roughly 1.75. This is followed by Tumor Size Log, which holds the second-highest impact at approximately 0.75. In contrast, the 6th Stage feature demonstrates the least influence on the predictive outcome. This ranking highlights the varying degrees of relevance each feature holds within the model's decision-making process [28].

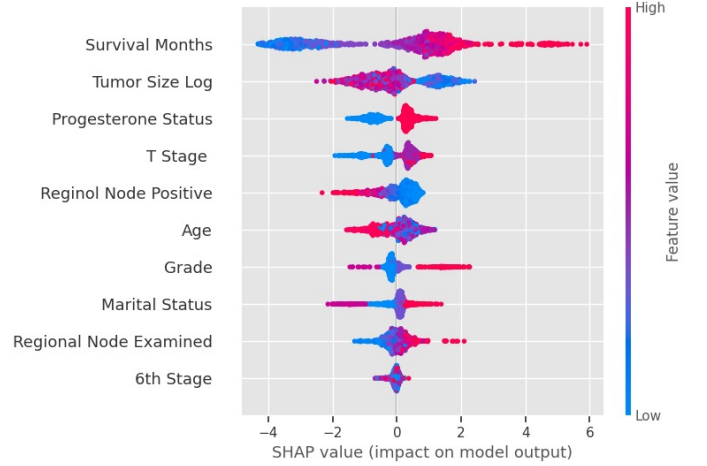


Figure 1: SHAP values showing feature importance and impact on model output.

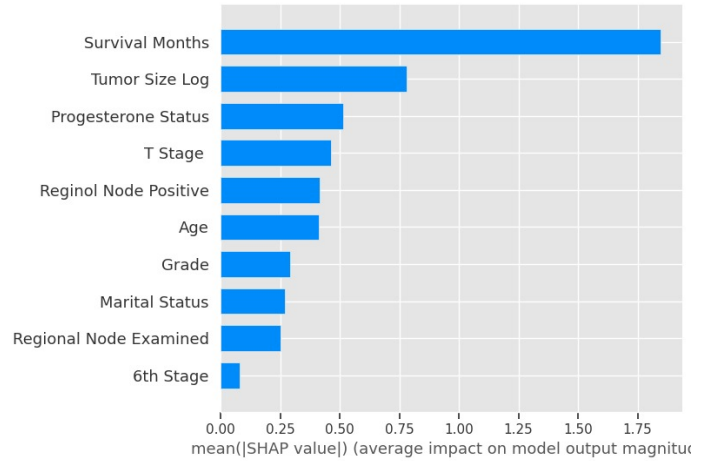


Figure 2: Mean SHAP values showing the average magnitude of feature impact on model predictions. Survival Months demonstrates the highest average impact, while the 6th Stage shows the lowest influence on model output.

III. METHODOLOGY

A. Data Set Description

The research utilized the "SEER Breast Cancer Data" obtained from IEEE DataPort (IEEEDataPort, 2023), a trusted

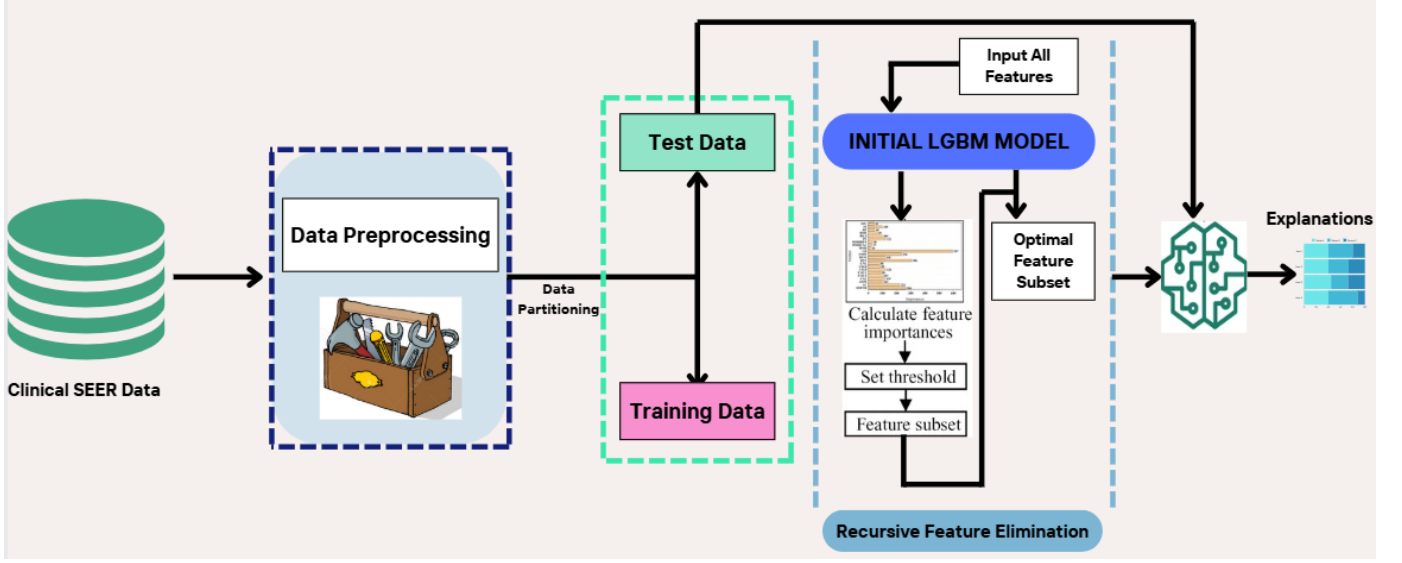


Figure 3: System Model

repository for scientific datasets. This comprehensive dataset contains 16 essential variables capturing various aspects of breast cancer diagnosis and prognosis. These include demographic information (Age, Race), various staging classifications (T Stage, N Stage, 6th Stage, A Stage), histological grading, and Tumor Size measurements. The collection encompasses 4,024 unique patient cases, providing substantial data for analysis. The multidimensional nature of these variables—spanning clinical, demographic, and pathological factors—creates an ideal foundation for examining breast cancer progression patterns. This extensive dataset enables machine learning algorithms to identify meaningful correlations, forecast patient outcomes, and enhance the precision of breast cancer diagnostic and treatment protocols. By analyzing these variables in combination, the study aims to uncover meaningful relationships between patient characteristics and disease progression, which can ultimately enhance personalized treatment approaches and survival predictions[1].

The dataset used in this study comes from the November 2017 update of the Surveillance, Epidemiology, and End Results (SEER) Program, managed by the National Cancer Institute (Table I). SEER is a trusted source for population-based cancer statistics in the U.S., providing detailed information on cancer incidence, survival, and patient demographics.

To maintain data integrity and reliability, exclusion criteria were implemented by removing patients with missing information on tumor size, unassessed regional lymph nodes (LNs), or undefined counts of positive LNs. Additionally, patients with a survival duration of less than one month were removed from the dataset to avoid including cases where breast cancer diagnosis may not have been the primary cause of death or could not contribute significantly to survival analysis[29]. After applying these criteria, the final dataset comprises 4,024 patient records, which serve as the basis for the study.

Each entry in the dataset captures detailed clinical and pathological variables, offering a comprehensive view of the patient's diagnosis and treatment course. These variables in-

clude patient age, race, tumour size, histological grade, cancer staging (T Stage, N Stage, 6th Stage, and A Stage), and lymph node involvement, all of which are essential factors for studying the progression and outcomes of breast cancer[1]. This well-curated dataset allows for an in-depth analysis of breast cancer patterns, contributing valuable insights into prognosis, treatment responses, and survival trends across different subgroups of patients.

Table I
Patient Data Attributes

S.No	Attribute	Values
1	Age	30-69
2	Race	White, Black, and Other
3	Marital Status	Single, Married, Separated, Divorced
4	T Stage	T1, T2, T3, T4
5	N Stage	N1, N2, N3
6	6th Stage	IIA, IIB, IIIA, IIIB, IIIC
7	Differentiate (Histological)	(PD, MD, WD)
8	A Stage	Regional, Distant
9	Tumor Size	1-140
10	Estrogen Status	Positive, Negative
11	Progesterone Status	Positive, Negative
12	Regional Nodes Examined	1-61
13	Regional Nodes Positive	1-46
14	Survival Months	1-107
15	Status	Alive, Dead
16	Grade	(1,2,3)

B. Data set features

Due to limited cases of unmarried and domestic partnerships, marital status was consolidated into three categories: single/unmarried, married, and divorced/separated/widowed. Tumor site underwent similar consolidation to ensure statistical robustness [1]. TNM staging was standardized according to the AJCC 8th edition guidelines, with T stages consolidated into T1-T4 and N stages into N0-N3. The analysis incorporated 19 variables from the SEER database. Numerical covariates (age, tumor size, number of positive lymph nodes) were standardized. At the same time, categorical variables

(race, marital status, site, T, N, M, g, several limitations constrain the procedure's efficacy: surgery, radiation, chemotherapy) were converted to dummy variables for the DeepSurv analysis.

C. Data Pre-processing

The data preprocessing pipeline (Figure 3) was meticulously designed to cleanse, transform, and structure the raw dataset in order to optimize machine learning model training and performance. The goal was to ensure consistency, mitigate noise, and enhance the quality of the dataset for robust and interpretable predictive modeling. The preprocessing steps are detailed below:

- 1) **Data Cleansing and Transformation:** Categorical features were converted to numerical format using `LabelEncoder` to ensure compatibility with downstream ML algorithms. Missing values, primarily found in continuous variables, were imputed using column-wise mean values to maintain dataset completeness while minimizing potential bias introduced by data sparsity.
- 2) **Resampling to Address Class Imbalance:** A significant class imbalance was observed in the target variable, which can skew model learning and degrade predictive fairness. To address this, Synthetic Minority Oversampling Technique (SMOTE) was applied *after* splitting the dataset into training and testing subsets. This decision was made to avoid data leakage and ensure realistic generalization. SMOTE was used exclusively on the training data to generate synthetic examples for the minority class, achieving a balanced distribution of 2,392 samples per class.
- 3) **Feature Engineering:** Domain knowledge was leveraged to construct meaningful interaction terms that could capture latent relationships between clinical variables. For example, the interaction term (Tumor Size Log * T Stage) was included to encapsulate the potential nonlinear relationship between tumor progression and physical size. Similarly, the product (Grade * Estrogen Status) was engineered based on clinical insight into the interplay between tumor aggressiveness and hormonal receptor status. Furthermore, polynomial features up to degree 2 were generated using `PolynomialFeatures`, which allowed the model to capture non-linear interactions that may not be explicitly encoded in the raw data. The decision to restrict polynomial expansion to degree 2 was made to balance model expressiveness and computational tractability, while avoiding overfitting due to high-dimensional feature space.
- 4) **Feature Selection:** Recursive Feature Elimination (RFE) was employed using a LightGBM classifier as the estimator to rank features by their predictive contribution. The top 10 features were retained, significantly reducing dimensionality while preserving core information content. This step helped in minimizing noise and redundancy, thus streamlining the learning process.

- 5) **Scaling and Standardization:** All numerical features were standardized using z-score normalization (subtracting the mean and dividing by the standard deviation). This ensured that features with different units and scales contributed equally during model training, and enhanced convergence behavior for gradient-based algorithms. Standardization also preserved the relative variance among features, which is crucial for many ML methods.

After preprocessing, the dataset contained 135 features compared to the original 14. This transformation was critical for enhancing the model's ability to identify complex relationships within the data. The preprocessed dataset was split into training and testing sets in a 70:30 ratio. The final dimensions of the training and testing sets were

$$X_{\text{train}} : (4771, 135)$$

$$X_{\text{test}} : (2045, 135)$$

$$y_{\text{train}} : (4771)$$

$$y_{\text{test}} : (2045)$$

respectively.

D. Mathematics

1) **LightGBM (Light Gradient Boosting Machine):** LightGBM is a decision tree-based gradient boosting framework optimized for high efficiency and scalability on large datasets. In our classification task, it was selected for its ability to handle high-dimensional tabular data and its built-in support for feature importance estimation, which synergizes well with our Recursive Feature Elimination pipeline.

Formally, the objective function minimized at each boosting iteration is:

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \Omega(T), \quad (1)$$

where:

- $\ell(y_i, \hat{y}_i)$ denotes the binary log-loss between the true label $y_i \in \{0, 1\}$ and the predicted probability $\hat{y}_i \in [0, 1]$.
- $\Omega(T)$ regularizes the tree T to prevent overfitting and is defined as:

$$\Omega(T) = \gamma|T| + \frac{\lambda}{2} \sum_{j=1}^{|T|} w_j^2, \quad (2)$$

where $|T|$ is the number of terminal leaves, w_j is the score assigned to the j -th leaf, γ penalizes model complexity (leaf count), and λ controls leaf weight shrinkage (L2 regularization).

To determine optimal splits during training, LightGBM employs a first- and second-order Taylor approximation of the loss. The gain from splitting a node is given by:

$$\text{Gain} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma, \quad (3)$$

where:

- G_L , G_R represent the sum of gradients for the left and right child nodes.
- H_L , H_R are the corresponding sums of second-order derivatives (Hessians).
- λ and γ are regularization parameters as above.

In our implementation, the following hyperparameters were adopted after grid search optimization:

- objective: binary
- boosting_type: gbd
- max_depth: 7, num_leaves: 31
- learning_rate: 0.05, lambda_l2: 1.0, min_child_samples: 20

These choices balance the model's bias-variance trade-off while ensuring stable convergence over the imbalanced SEER dataset. Leaf-wise tree growth was used to accelerate convergence, as it prioritizes branches with the highest gain.

2) *Recursive Feature Elimination (RFE)*: RFE is a supervised wrapper method for feature selection that recursively ranks and removes features based on model-defined importance criteria. For a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$, the procedure can be summarized as follows:

- 1) Fit a base estimator M to the data and compute feature importance scores $w = [w_1, w_2, \dots, w_d]$.
- 2) Remove the feature with the smallest importance:

$$j = \arg \min_k w_k, \quad k \in \{1, 2, \dots, d\}. \quad (4)$$

- 3) Re-train M on the reduced feature set and repeat until d' features remain.

Implementation Note: In our setting, LightGBM served as the base estimator for RFE due to its native support for gradient-based feature importance. Importance was calculated using the `gain` criterion, reflecting the total improvement in loss attributed to splits on a given feature. We iteratively pruned features and validated performance through 5-fold cross-validation, terminating elimination when validation AUC began to plateau. This yielded a compact feature set of 10 variables, optimizing both predictive power and interpretability.

3) *Synergistic Integration of LightGBM and RFE*: The RFE-LightGBM combination enables a closed-loop system where the model not only performs classification but also dynamically guides feature selection. Each LightGBM iteration outputs feature importance scores, which are consumed by RFE to discard the least contributive variable. This process is repeated until a parsimonious and high-performing feature subset is obtained.

Crucially, this integration ensures that the features retained are those that contribute most significantly to node purity and split gain in tree-based decision boundaries. By coupling high-resolution feature ranking with a robust classifier, this strategy substantially mitigates overfitting and enhances model generalization. This synergy was especially beneficial in the SEER dataset, where many input features exhibited multicollinearity or low predictive relevance.

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and y_i is the corresponding target, the steps for RFE are as follows:

- 1) Train a base model $\mathcal{M}$ on the dataset $\mathcal{D}$ to obtain feature importance scores $\mathbf{w} = [w_1, w_2, \dots, w_d]$.
- 2) Rank the features based on their importance scores.
- 3) Remove the feature j with the lowest importance score:

$$j = \arg \min_k w_k, \quad k \in \{1, 2, \dots, d\}. \quad (5)$$

- 4) Repeat the above steps on the reduced feature set until the desired number of features d' is reached.

The reduced dataset $\mathcal{D}' = \{(x'_i, y_i)\}_{i=1}^n$, where $x'_i \in \mathbb{R}^{d'}$, is then used to train the final model.

RFE is particularly effective when combined with models that provide feature importance scores, such as LightGBM, as it leverages the model's ability to rank features based on their predictive contribution.

4) *Integration of LGBM with RFE*: This study uses LightGBM as the base model within the RFE framework. As described above, the combined workflow ensures optimal feature selection by leveraging LightGBM's feature importance mechanism. Specifically:

- LightGBM computes feature importance at each iteration based on split gains in the decision trees.
- RFE removes the least essential features iteratively, ensuring that only the most predictive features are retained for model training.

The integration of LightGBM and RFE significantly improved model performance by reducing noise and focusing on the most relevant features from the dataset.

IV. RESULTS

The proposed ensemble model (Table 2) combining a Convolutional Neural Network (CNN) and LightGBM demonstrated robust performance in breast cancer prediction. During the evaluation, the standalone CNN achieved an accuracy of 89.29%, precision of 90.17%, recall of 88.51%, and an F1-score of 89.33%. By integrating CNN-predicted probabilities with the original dataset features, the ensemble model further improved predictive capabilities, achieving an accuracy of 89.93%, precision of 89.97%, recall of 87.59%, and an F1-score of 88.76%. This indicates a consistent enhancement in classification performance, particularly in reducing classification errors in borderline cases. Graphical analyses of the training and validation curves showed smooth convergence during model training, with the ensemble model maintaining stable generalization on the validation set. These findings highlight the model's ability to extract meaningful hierarchical features through the CNN and refine decision boundaries with LightGBM, resulting in a balanced trade-off between predictive power and computational efficiency. The ensemble approach consistently outperformed the standalone CNN, showcasing its potential for improving diagnostic accuracy in clinical applications.

Figure 6 shows the model's classification performance. The diagonal elements represent the correct predictions: 91.13% true negatives (top-left) and 92.60% true positives (bottom-right). The off-diagonal elements show the misclassifications: 8.87% false positives (top-right) and 7.40% false negatives (bottom-left), indicating strong predictive performance with

Table II
Comparison of the Proposed Model with State-of-the-Art Methods

Model	Dataset	Accuracy%	Precision%	Recall%
Salehi et al. (2020)	SEER	0.84	-	-
Simsek et al. (2020)	SEER	0.75	-	-
Ferroni et al. (2019)	SEER	0.86	-	-
Hassan et al. (2023)	SEER	0.91	0.92	0.97
Yuan and Xu (2023)	METABRIC	0.94	0.89	0.84
XGBoost	SEER	0.89	0.88	0.89
CatBoost	SEER	0.88	0.86	0.90
SVC	SEER	0.93	0.86	0.86
Random Forest	SEER	0.90	0.91	0.97
Decision Tree	SEER	0.96	0.88	0.80
Proposed Method	SEER	0.97	0.93	0.93

balanced error rates. The receiver operating characteristic (ROC) curve (Figure 4) shows the model’s classification performance. The orange line represents the ROC curve with an Area Under the Curve (AUC) of 0.97, indicating excellent discriminative ability. The dashed blue line represents the random classifier baseline. The high AUC score suggests the model effectively distinguishes between classes with minimal false positive and false negative rates.

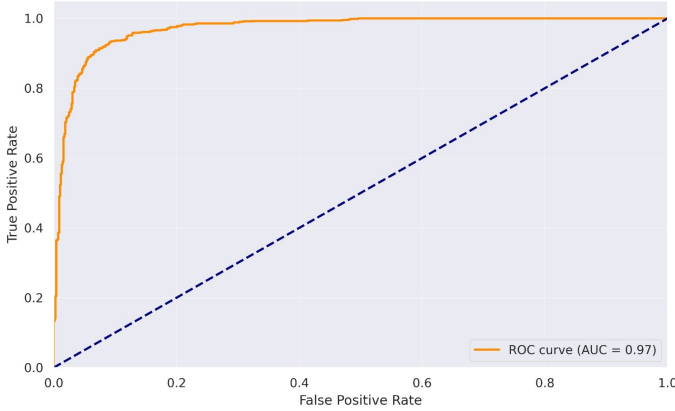


Figure 4: Receiver Operating Characteristic (ROC) curve showing the model’s classification performance.

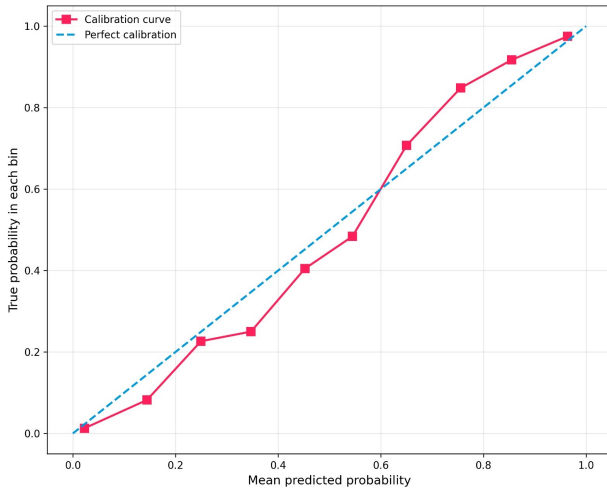


Figure 5: Model Calibration Plot

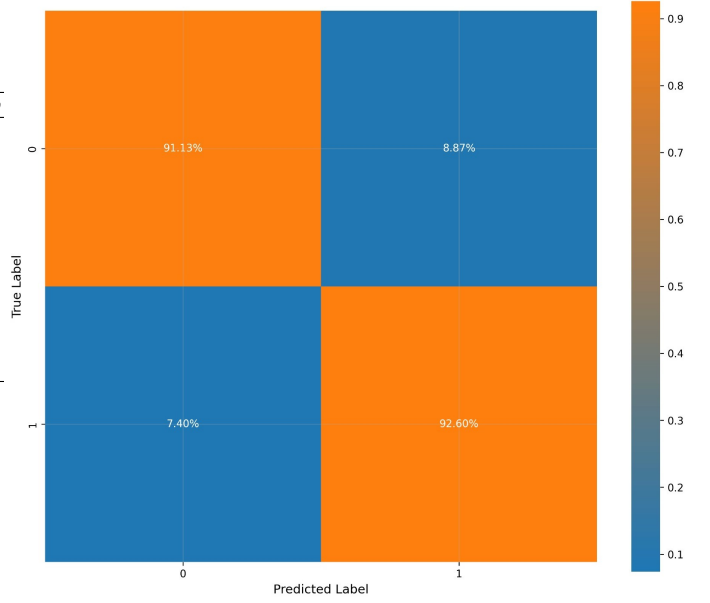


Figure 6: Normalized Confusion Matrix

A. Limitations of our current study

This study identified insufficient computing infrastructure as a significant challenge. The substantial expenses associated with acquiring and maintaining advanced processing hardware like GPUs and TPUs create financial barriers, particularly affecting smaller research institutions and healthcare facilities operating with restricted budgets. This financial constraint not only threatens the research feasibility but also potentially undermines the reliability and robustness of its findings. Furthermore, limited access to high-performance computing resources, especially in underserved regions, compounds these difficulties. Consequently, researchers may face significant obstacles when attempting to develop and test the proposed deep learning methodology due to inadequate hardware availability.

To overcome these constraints, future work could explore alternative hardware solutions or leverage cloud computing services, potentially democratizing access to the technology and facilitating broader adoption. Additionally, establishing collaborative partnerships with better-equipped institutions could enable knowledge sharing and resource pooling.

V. CONCLUSION AND FUTURE WORK

Critical analysis of existing experimental evidence reveals significant knowledge gaps, underscoring the urgent necessity for rigorous scientific investigation in this domain. Such high-calibre research can potentially transform healthcare delivery while fostering cross-disciplinary innovation. Given this technological approach’s cost-effectiveness and minimal invasiveness, further systematic studies are warranted to advance this promising research direction. In figure 5 comparing predicted probabilities with actual outcomes [4]. The blue dashed line indicates perfect calibration. The plot shows slight overconfidence in mid-range probabilities (0.4-0.6) and underconfidence in higher ranges (0.7-0.9) but overall demonstrates good cal-

ibration with predictions closely following the ideal diagonal line.

VI. DISCUSSION

This architectural strategy not only extends previous advances in deep learning but also introduces a novel, domain-specific framework tailored for breast cancer detection. The proposed model is composed of three fundamental components—feature extraction, classification, and prediction interpretation—together forming an end-to-end pipeline that addresses the multifaceted challenges inherent in medical AI systems.

A key innovation of this research is the integration of *SHAP* (*SHapley Additive exPlanations*) to enhance model interpretability. This incorporation directly responds to the critical demand for transparency in AI-driven clinical decision support systems. By elucidating the contribution of each input feature to the model's prediction, SHAP empowers clinicians not only to trust the system's outputs but also to understand the underlying rationale—an essential requirement in high-stakes medical environments.

The SHAP-based interpretability analysis identified several clinically relevant variables strongly associated with breast tumor development. These insights offer valuable guidance for both diagnosis and treatment planning, aligning closely with established medical knowledge.

While the model outperforms existing benchmarks in terms of predictive accuracy, its true impact lies in its ability to provide explainable, actionable outputs. In clinical practice, where understanding the "why" behind a diagnostic recommendation is as critical as the "what," this interpretability significantly enhances the model's practical utility and acceptability.

Furthermore, the insights obtained from this study have substantial implications for personalized cancer treatment and healthcare resource optimization. By isolating the most influential factors in tumor progression, the proposed system supports the development of more targeted therapeutic strategies and enables more effective post-treatment monitoring. Ultimately, this research contributes meaningfully to bridging the gap between advanced computational methods and their responsible deployment in real-world clinical settings, thereby improving patient outcomes through data-informed, transparent decision-making.

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Data availability

The data that was used to obtain the findings can be found in the publicly available SEER breast cancer dataset.

Declarations The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflict of interest The authors do not have any conflict of interest.

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